Interop QOS

2026-03-11-16-14-08

Objective

The objective of the QoS (Quality of Service) traffic throughput test is to measure the maximum achievable throughput of a network under specific QoS settings and conditions.By conducting this test, we aim to assess the capacity of network to handle high volumes of traffic while maintaining acceptable performance levels,ensuring that the network meets the required QoS standards and can adequately support the expected user demands.

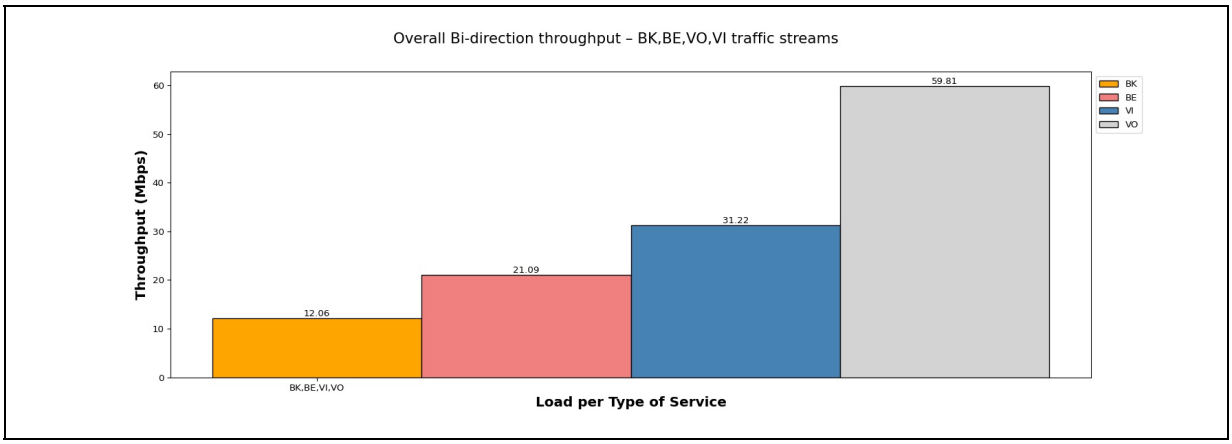
Test Configuration	Device List	DESKTOP-DDPI3HE(Windows), oppophone(Android), sungsung(Android), macoss-MacBook-Air.local(Mac)
	Number of Stations	Total(4) Android(2) Windows(1) Mac(1)
	AP Model	NETGEAR
	SSID	NETGEAR_2G_wpa2
	Traffic Duration in hours	0.02
	Security	wpa2
	Protocol	TCP
	Traffic Direction	Bi-direction
	TOS	['VI', 'VO', 'BE', 'BK']
	Per TOS Load in Mbps	Upload:100.0 Mbps,Download:100.0 Mbps

Overall Bi-direction Throughput for all TOS i.e BK | BE | Video (VI) | Voice (VO)

No of Stations	Throughput for Load 100.0 Mbps-upload	Throughput for Load 100.0 Mbps-download
4	BK : 5.67, BE : 15.87, VI: 19.6, VO: 37.06	BK : 6.39, BE : 5.22, VI: 11.62, VO: 22.75

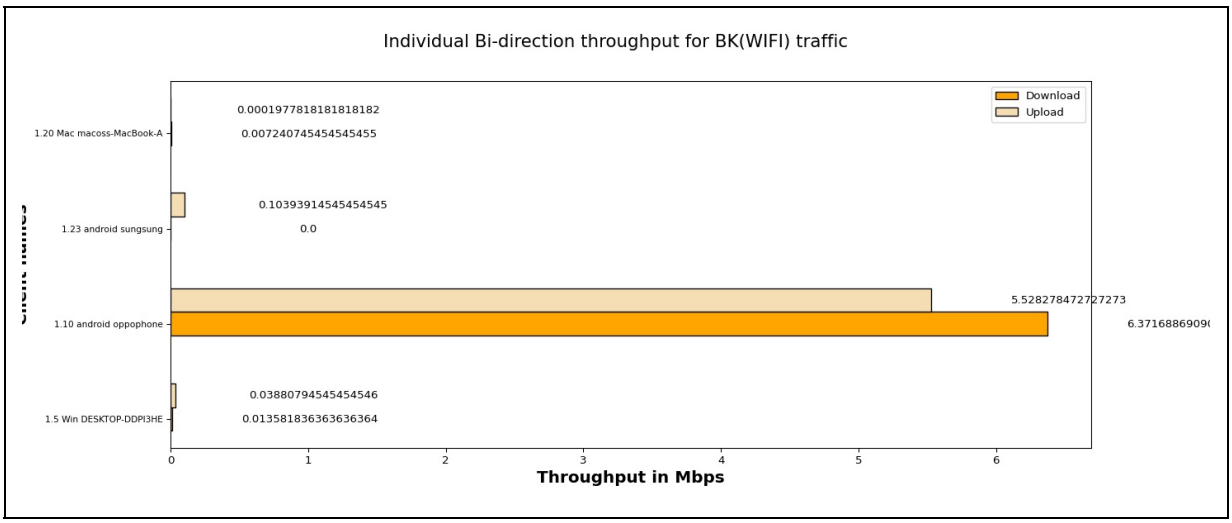
Overall Bi-direction throughput for 4 clients with different TOS.

The below graph represents overall Bi-direction throughput for all connected stations running BK, BE, VO, VI traffic with different intended loadsUpload:100.0 Mbps,Download:100.0 Mbps per tos



Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic BK(WiFi).

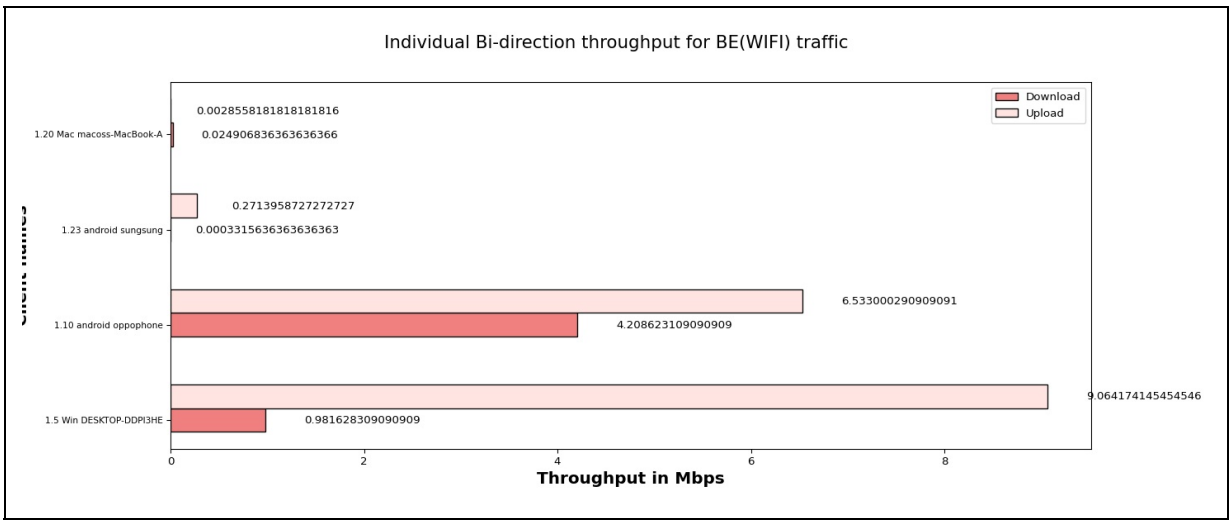
The below graph represents individual throughput for 4 clients running BK (WiFi) traffic. X- axis shows “Throughput in Mbps” and Y-axis shows “number of clients”.



Client Name	MAC	SSID	Type of traffic	Offered upload rate	Offered download rate	Observed average upload rate	Observed average download rate	Observed Upload Drop (%)	Observed Download Drop (%)	Expected Bi-direction rate(Mbps)	Status
1.5 Win DESKTOP-DDPI3HE	e4:a4:71:93:70:e7	NETGEAR_2G_wpa2	Background	100.0 Mbps	100.0 Mbps	0.03880794545454546 Mbps	0.013581836363636364 Mbps	45.303030	78.787879	30	FAIL
1.10 android oppophone	e6:5b:3a:c3:48:ed	NETGEAR_2G_Open	Background	100.0 Mbps	100.0 Mbps	5.528278472727273 Mbps	6.371688690909091 Mbps	13.004777	16.171715	30	FAIL
1.23 android sungsung	46:ca:12:8c:58:67	NETGEAR_2G_wpa2	Background	100.0 Mbps	100.0 Mbps	0.10393914545454545 Mbps	0.0 Mbps	8.522727	61.818182	30	FAIL
1.20 Mac macoss-MacBook-Air.local	30:35:ad:c5:e1:be	NETGEAR_2G_wpa2	Background	100.0 Mbps	100.0 Mbps	0.0001977818181818182 Mbps	0.007240745454545455 Mbps	87.272727	54.545455	30	FAIL

Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic BE(WiFi).

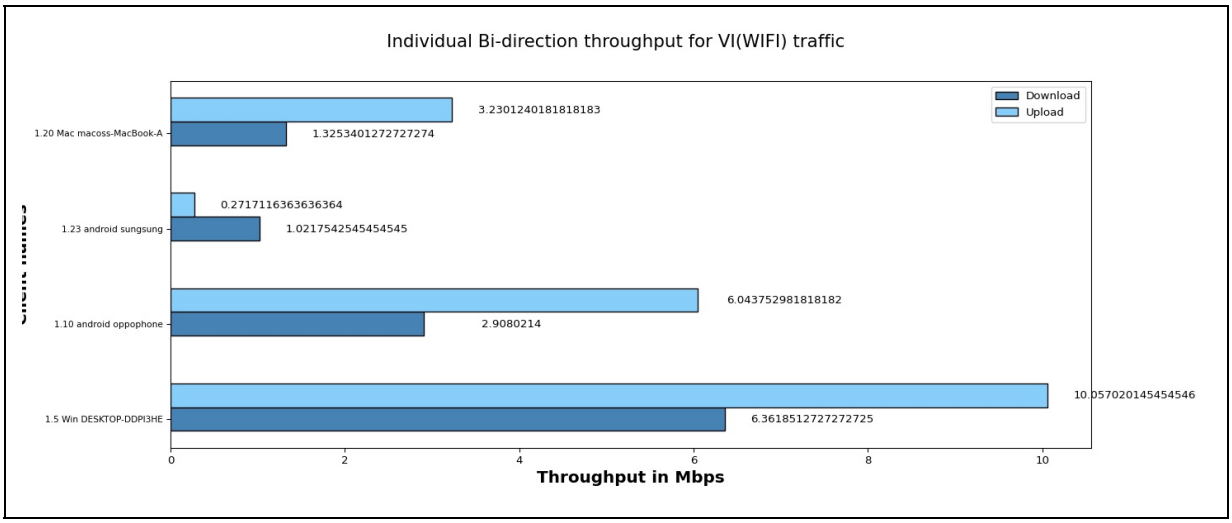
The below graph represents individual throughput for 4 clients running BE (WiFi) traffic. X- axis shows “number of clients” and Y-axis shows “Throughput in Mbps”.



Client Name	MAC	SSID	Type of traffic	Offered upload rate	Offered download rate	Observed average upload rate	Observed average download rate	Observed Upload Drop (%)	Observed Download Drop (%)	Expected Bi-direction rate(Mbps)	Status
1.5 Win DESKTOP-DDPI3HE	e4:a4:71:93:70:e7	NETGEAR_2G_wpa2	Besteffort	100.0 Mbps	100.0 Mbps	9.064174145454546 Mbps	0.981628309090909 Mbps	3.923550	57.792365	30	FAIL
1.10 android oppophone	e6:5b:3a:c3:48:ed	NETGEAR_2G_Open	Besteffort	100.0 Mbps	100.0 Mbps	6.533000290909091 Mbps	4.208623109090909 Mbps	13.451325	17.787438	30	FAIL
1.23 android sungsung	46:ca:12:8c:58:67	NETGEAR_2G_wpa2	Besteffort	100.0 Mbps	100.0 Mbps	0.2713958727272727 Mbps	0.0003315636363636363 Mbps	79.103367	99.191919	30	FAIL
1.20 Mac macoss-MacBook-Air.local	30:35:ad:c5:e1:be	NETGEAR_2G_wpa2	Besteffort	100.0 Mbps	100.0 Mbps	0.0028558181818181816 Mbps	0.024906836363636366 Mbps	100.000000	63.030303	30	FAIL

Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic VI(WiFi).

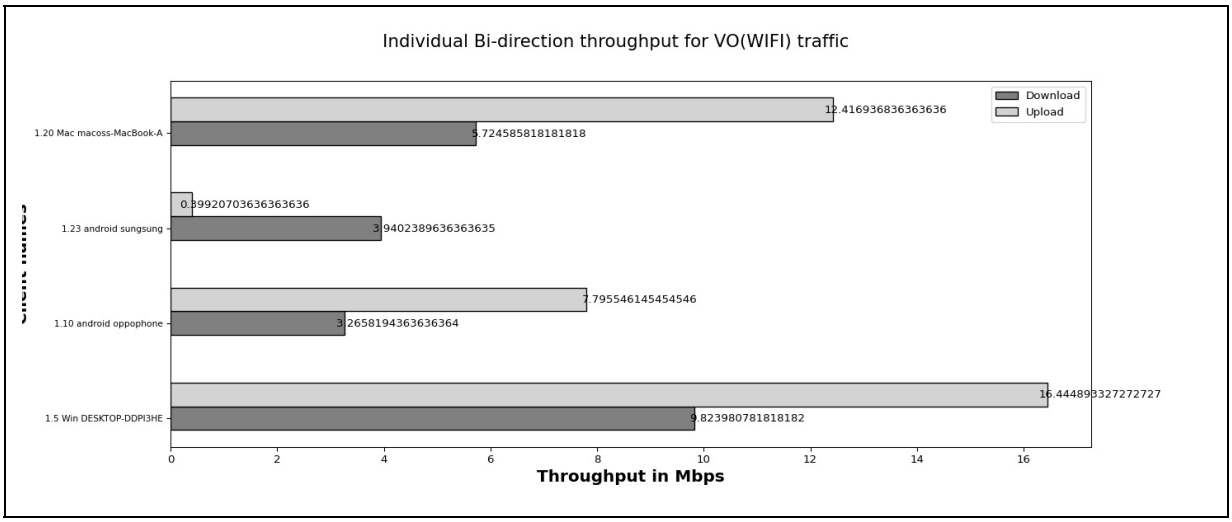
The below graph represents individual throughput for 4 clients running VI (WiFi) traffic. X- axis shows "number of clients" and Y-axis shows "Throughput in Mbps".



Client Name	MAC	SSID	Type of traffic	Offered upload rate	Offered download rate	Observed average upload rate	Observed average download rate	Observed Upload Drop (%)	Observed Download Drop (%)	Expected Bi-direction rate(Mbps)	Status
1.5 Win DESKTOP-DDPI3HE	e4:a4:71:93:70:e7	NETGEAR_2G_wpa2	Video	100.0 Mbps	100.0 Mbps	10.057020145454546 Mbps	6.3618512727272725 Mbps	0.145587	11.418664	30	FAIL
1.10 android oppophone	e6:5b:3a:c3:48:ed	NETGEAR_2G_Open	Video	100.0 Mbps	100.0 Mbps	6.043752981818182 Mbps	2.9080214 Mbps	7.210580	6.399471	30	FAIL
1.23 android sungsung	46:ca:12:8c:58:67	NETGEAR_2G_wpa2	Video	100.0 Mbps	100.0 Mbps	0.2717116363636364 Mbps	1.0217542545454545 Mbps	31.809703	24.225324	30	FAIL
1.20 Mac macoss-MacBook-Air.local	30:35:ad:c5:e1:be	NETGEAR_2G_wpa2	Video	100.0 Mbps	100.0 Mbps	3.2301240181818183 Mbps	1.3253401272727274 Mbps	0.842355	23.425415	30	FAIL

Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic VO(WiFi).

The below graph represents individual throughput for 4 clients running VO (WiFi) traffic. X- axis shows "number of clients" and Y-axis shows "Throughput in Mbps".



Client Name	MAC	SSID	Type of traffic	Offered upload rate	Offered download rate	Observed average upload rate	Observed average download rate	Observed Upload Drop (%)	Observed Download Drop (%)	Expected Bi-direction rate(Mbps)	Status
1.5 Win DESKTOP-DDPI3HE	e4:a4:71:93:70:e7	NETGEAR_2G_wpa2	Voice	100.0 Mbps	100.0 Mbps	16.444893327272727 Mbps	9.823980781818182 Mbps	0.010674	6.129476	30	FAIL
1.10 android oppophone	e6:5b:3a:c3:48:ed	NETGEAR_2G_Open	Voice	100.0 Mbps	100.0 Mbps	7.795546145454546 Mbps	3.2658194363636364 Mbps	9.678345	33.224774	30	FAIL
1.23 android sungsung	46:ca:12:8c:58:67	NETGEAR_2G_wpa2	Voice	100.0 Mbps	100.0 Mbps	0.39920703636363636 Mbps	3.9402389636363635 Mbps	48.280048	48.353352	30	FAIL
1.20 Mac macoss-MacBook-Air.local	30:35:ad:c5:e1:be	NETGEAR_2G_wpa2	Voice	100.0 Mbps	100.0 Mbps	12.416936836363636 Mbps	5.724585818181818 Mbps	2.488391	30.168760	30	FAIL

Information	contact	support@candelatech.com